Laboratory Environments Mask Gait Differences Between Healthy Participants and Patients With Anterior Cruciate Ligament Reconstruction

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***Abstract*— Previous work suggests that people with mobility impairments move differently in the traditional gait laboratory than in natural environments, but the extent to which this possible divergence may have limited understanding of gait following anterior cruciate ligament reconstruction (ACLR) remains unknown. We hypothesized that 1) patients following ACLR walk more asymmetrically in daily life than in the laboratory and 2) differences between patients following ACLR and individuals with no gait pathologies would be more pronounced in daily life than in the laboratory. Twelve participants (six post-ACLR, six healthy) completed gait assessments in the laboratory with optical motion capture and inertial measurement units (IMUs) and in daily life with IMUs. Patients walked similarly to healthy participants in the laboratory, but in daily life they walked with a longer gait-cycle time (1.22 ± 0.03 s vs. 1.08 ± 0.06 s), longer double-support phase (21.8 ± 1.3 % vs. 17.9 ± 2.7 % Gait Cycle Time), and greater single-support asymmetry (8.4 ± 0.9 % vs 4.1 ± 0.7 %). While the reasons behind the observed differences were not studied, these results suggest that gait-analysis studies to date may have not realistically captured natural-environment behavior. Wearable sensors now offer a path toward deeper understanding of post-surgical gait and the specific patterns that may place certain patients at risk for post-traumatic osteoarthritis.**

***Index Terms*— Motion capture, gait, inertial measurement units, anterior cruciate ligament reconstruction.**

I. INTRODUCTION

THE laboratory environment has been the primary setting used to study the link between gait abnormalities following anterior cruciate ligament reconstruction (ACLR) and risk for post-traumatic osteoarthritis (PTOA), but this approach has notable limitations. Marker-based motion capture, which has been historically used to study gait, requires designated laboratory space, expensive equipment, and trained biomechanists, preventing scalability to clinics and large biomechanics research studies. Additionally, the presence of researchers may influence how patients move. The white coat syndrome is a well-described phenomenon in vascular research wherein an individual's blood pressure rises in the presence of a physician [1]. Similarly, the Hawthorne effect—a phenomenon where study participants modify their behavior due to an awareness of being watched—has also been observed in other fields [2]. Recent work has identified a similar effect in gait analysis, where study participants increase their gait speed, cadence, and stride length when researchers are present in the laboratory [3], [4], [5]. Patients with a femoral prosthesis, as well as patients following stroke, were also found to walk more asymmetrically when they were not aware that they were being watched [6], [7]. In further support of this premise, older adults and patients with Parkinson's disease and multiple sclerosis have been found to walk with greater variability, greater asymmetry, and lower pace in natural environments compared to the laboratory [8], [9], [10], [11], [12].

Although wearable sensors, such as inertial measurement units (IMUs), are emerging as promising, cost-effective technologies for gait assessment, studies deploying these tools for monitoring of orthopaedic patients during daily life are limited [13], [14], [15]. Remote monitoring requires patient compliance and produces large amounts of data. While prior epidemiology studies, such as the Osteoarthritis Initiative [16], have documented feasibility in thousands of patients, activity monitoring with a belt-worn or wrist-worn accelerometer, or even smartwatch [17], is limiting. Following orthopedic surgery, such as ACLR, lower-limb kinematics and kinetics are of interest [18], [19], [20], but these outcomes require data from sensors placed on multiple body segments, which is burdensome for patients. Widespread adoption of multi-sensor protocols needs to be preceded by compelling evidence that any potential burden on patients and researchers will be eclipsed by the insight we can gain. As rigid and bulky sensors are replaced with more flexible and compliant ones, such as band-aid and even tattoo-like sensors [21], the burden on patients will subside [22], but not disappear. As hardware becomes unobtrusive, evidence on the potential of wearables to transform understanding of gait pathologies is needed to support their uptake for remote patient monitoring.

Existing knowledge of post-ACLR gait is largely based on laboratory assessments, with no study to date providing evidence of how daily-life monitoring may reshape these insights. We posit that under the watchful eye of researchers most participants move the way they "should" [3], but that in daily life patients after ACLR may be more inclined than healthy participants to adopt pain-avoidance strategies,

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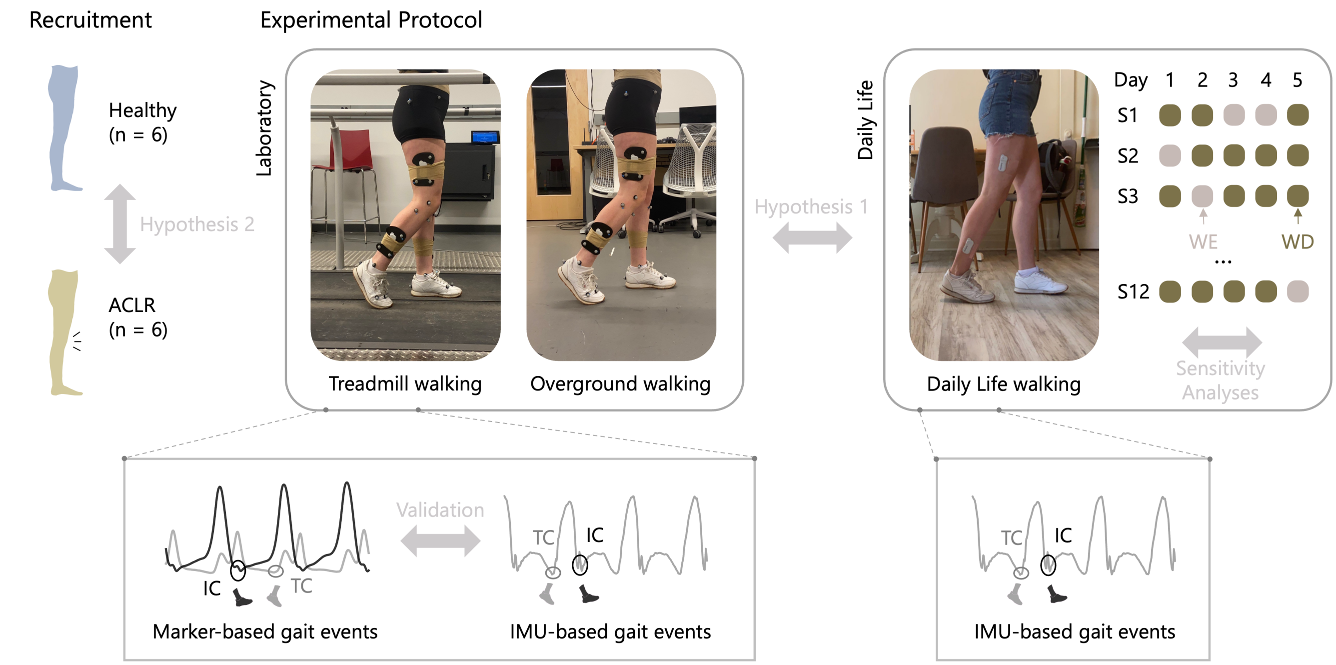

Fig. 1. Experimental Setup. Six patients post-ACLR and six healthy participants were recruited for this study. Participants wore four IMUs, one on each thigh and shank, during remote monitoring (right), for at least five days. They were outfitted with optical motion capture markers and four IMUs for overground (middle) and treadmill (left) walking in the laboratory. Initial contact (IC) and terminal contact (TC) were identified in all IMU data and validated against lab-based marker outcomes. Hypothesis 1 studied how ACLR patient gait changed in each environment. Hypothesis 2 studied differences between healthy and ACLR patient groups in each environment. Additional sensitivity analyses exploring the influence of number of days of data and whether the data were collected on a weekday (WD) or weekend (WE) for each subject (S) on gait outcomes were completed.

eliciting greater detectable differences between the two populations. Accordingly, the aim of this study was to investigate whether laboratory environments cloud the conclusions we draw about post-surgical gait, providing initial support for the move toward natural-environment biomechanics. We hypothesized that patients after ACLR walk with a longer gait-cycle time, greater reliance on double support, and a more pronounced limp in daily life compared to the laboratory. Additionally, we hypothesized that differences between patients after ACLR and healthy participants would be higher in daily life than in the laboratory.

II. Methods

Six patients post-ACLR (3M, 3F; mean ± std age: 16.83 ± 2.48 years; height: 1.74 ± 0.05 m; weight: 68.175 ± 7.82 kg) and six healthy participants (3M, 3F; mean ± std age: 22.83 ± 2.78 years; 1.76 ± 0.078 m; 66.0 ± 9.2 kg) were recruited for this study, after receiving approval from Carnegie Mellon University's Institutional Review Board and obtaining informed consent. Initially, eight patients and seven healthy participants were assessed for eligibility. Two patients failed to follow up and one healthy participant only completed three days of remote monitoring data. Therefore, these three participants were removed from the study before data analysis, yielding a final sample size of 12 participants. Six ACLR patients underwent a primary single-bundle reconstruction with an autologous quadriceps tendon on a single limb by the same surgeon. One of those patients had undergone an ACLR on their contralateral limb two years before their most recent surgery. To be eligible to participate, healthy participants had to have no prior history of knee injury or any other major musculoskeletal injury. ACLR patients had to be within 10 weeks of injury at the time of surgery. We recorded their gait three months after the surgery.

We evaluated walking in three conditions: laboratory treadmill, laboratory overground, and daily life (Figure 1). In the laboratory, participants were tracked with marker-based motion capture and IMUs for two trials of treadmill walking (2 minutes per trial) and two trials of overground walking (30 seconds per trial). We measured each participant's comfortable walking pace during an overground trial. This pace was then used during the treadmill trial. Participants were outfitted with a 41-marker modified Rizzoli markerset [23]. Marker-based motion capture data were collected at 100 Hz using a 20-camera motion capture system (OptiTrack, NaturalPoint Inc., Corvallis, OR). Simultaneously, IMU data were collected at 125 Hz with four flexible IMU sensors (MC10 Biostamp, Boston, MA). Each IMU was placed laterally on the thigh and shank of each limb. These six-axis IMUs recorded linear acceleration and angular velocity data across three degrees of freedom. Participants completed a series of three hops before each trial to synchronize marker and inertial data. For the daily life monitoring component, participants were sent home with four IMU sensors and instructed to wear them for five consecutive days, with the sensors placed laterally on the thighs and shanks. Only the shank IMUs were used to derive the outcomes reported here.

Temporal parameters of gait were extracted from marker-based motion capture collected in the laboratory and IMU data collected in each environment. Marker data were filtered with a fourth-order high-pass filter with a cut-off frequency of 15 Hz. To segment gait cycles, we used terminal contact, defined

TABLE I
SUMMARY OF TEMPORAL PARAMETERS

Parameter	Formula
Gait Cycle Time (GCT) [25]	$IC_{Target+1} - IC_{Target}$
Double Support [25]	$\frac{(TC_{Adjacent} - IC_{Target}) + (TC_{Target} - IC_{Adjacent})}{GCT} \times 100$
Limp [25]	$\frac{\lvert(TC_{Adjacent} - IC_{Target}) - (TC_{Target} - IC_{Adjacent})\rvert}{GCT} \times 100$
Swing Time [25]	$\frac{IC_{Target+1} - TC_{Target}}{GCT} \times 100$
Single Support [25]	$\frac{(IC_{Adjacent} - TC_{Adjacent})}{GCT} \times 100$
Step Time Asymmetry [29]	$Step\ Time\ (ST) = \frac{(IC_{Adjacent} - TC_{Target})}{GCT} \times 100$ $\lvert \ln \frac{Short\ ST}{Long\ ST} \rvert \times 100$
Single Support Asymmetry [29]	$\lvert \ln \frac{Short\ Single\ Support}{Long\ Single\ Support} \rvert \times 100$

Seven temporal parameters were assessed to characterize gait in laboratory and daily life conditions. Detection of gait events, such as initial contact (IC) and terminal contact (TC), from angular velocity data and a-priori designation of the dominant/reconstruction limb (target) and contralateral one (adjacent) informed estimation of these parameters. Primary outcomes are denoted in bold. After computing step time (ST) and single support time for each leg, the log of the ratio of the shorter to the longer parameter in a gait cycle was used tocalculate asymmetry.

as the moment the gradient of position surpasses 0.2 m/s following minimum toe-marker position, and initial contact, defined as the minimum heel-marker position after swing. IMU data were downsampled to 100 Hz to enable comparisons with the motion capture data. Angular velocities from the gyroscopes were filtered with a high-pass filter with a cutoff frequency of 0.25 Hz, followed by a low-pass filter with a cut-off frequency of 35 Hz. These filtering hyperparameters have been previously shown to minimize event identification delay and sensor drift [24], [25]. For the daily-life data, we identified walking segments using sagittal shank angular velocity thresholds of 2 to 10 rad/s for swing peaks and -2 to 2 rad/s for stance peaks. These thresholds were determined using a sensitivity analysis. Only stance peaks that were immediately followed by a swing peak were kept for analysis and grouped into bouts of continuous gait cycles. Walking bouts with five or more gait cycles were selected for analysis. The first and last two cycles were removed from the laboratory overground and each daily-life walking bout to allow each participant to reach a steady-state when walking [26], [27], [28]. The first and last 30 cycles were removed from the treadmill trial to allow the participants to adapt to the treadmill speed [20]. To identify steps from the IMU data, we used angular velocities from the shank sensors, first identifying swing peaks in the sagittal plane and then identifying terminal contact and initial contact events [25]. Once steps were identified, seven temporal parameters, computed using previously reported approaches [25], [29], were estimated (Table I).

Repeated measures (RM) analyses of variance (ANOVA) were used to test our two leading hypotheses, with the environment as our repeated measure. A one-way RM ANOVA was used to test the first hypothesis that patients post-ACLR walk differently in daily life compared to the two laboratory conditions. Post-hoc paired t-tests were used to determine individual differences between condition pairs (treadmill, overground, daily life). To test the second hypothesis that differences between ACLR patients and healthy participants are more pronounced in daily life than the laboratory, we used a two-way RM ANOVA with the two factors representing environment and health status (ACLR, healthy). A Shapiro-Wilk test, applied to the model residuals, was used to test the normality assumption. Post-hoc unpaired t-tests were used to determine individual differences between health status groups (ACLR, healthy) in each environment. For all the tests, significance was set at 0.017, after applying a Bonferroni correction for the three primary outcomes used to test the hypotheses.

To place the reported effect sizes in context with measurement accuracy, we report the errors of IMU-derived parameters relative to marker-based ones. For any IMU-derived differences between groups and across environments that were statistically significant, we note whether these differences are also greater than the accuracy of the IMU-derived parameters. When differences were not statistically significant (i.e. the null hypothesis could not be rejected), we carried out post-hoc power analyses to determine the sample sizes necessary to identify differences, should they exist.

For the daily-life outcomes, we performed sensitivity analyses with three different linear mixed effects models to determine the effect of the day type (weekday vs. weekend), number of days participants are monitored, and walking-bout length on gait outcomes. We found this sensitivity analysis to be important given the compliance issues often encountered in remote-monitoring studies. Random effects included the subject, while fixed effects included group (Healthy vs ACLR) and one additional study factor: day type, number of days monitored, or walking-bout length, taking the following general form: Outcome ~ Group * Study Factor + (1|Subject). Independent models were fit for each outcome and study factor because these factors operate at different hierarchical levels and could not be simultaneously represented within a single mixed-effects model structure. First, a linear mixed effects model with fixed effects for group and number of days of monitoring was fit to explore the influence of the number of days included in analysis on outcomes. Two participants (1 ACLR and 1 Healthy) only completed four, instead of five, days of data collection due to sensor failure. Therefore, we tested a range of 1 – 5 days to ensure that findings are insensitive to variability in participant monitoring time. Second, a linear mixed effects model with fixed effects for group and day type was used to determine the influence of day type. Next, a linear mixed-effects model, with group and walking-bout length serving as fixed effects, was used to assess how these outcomes changed with walking-bout length. Walking bouts were grouped into clusters by bout length including 5-10,

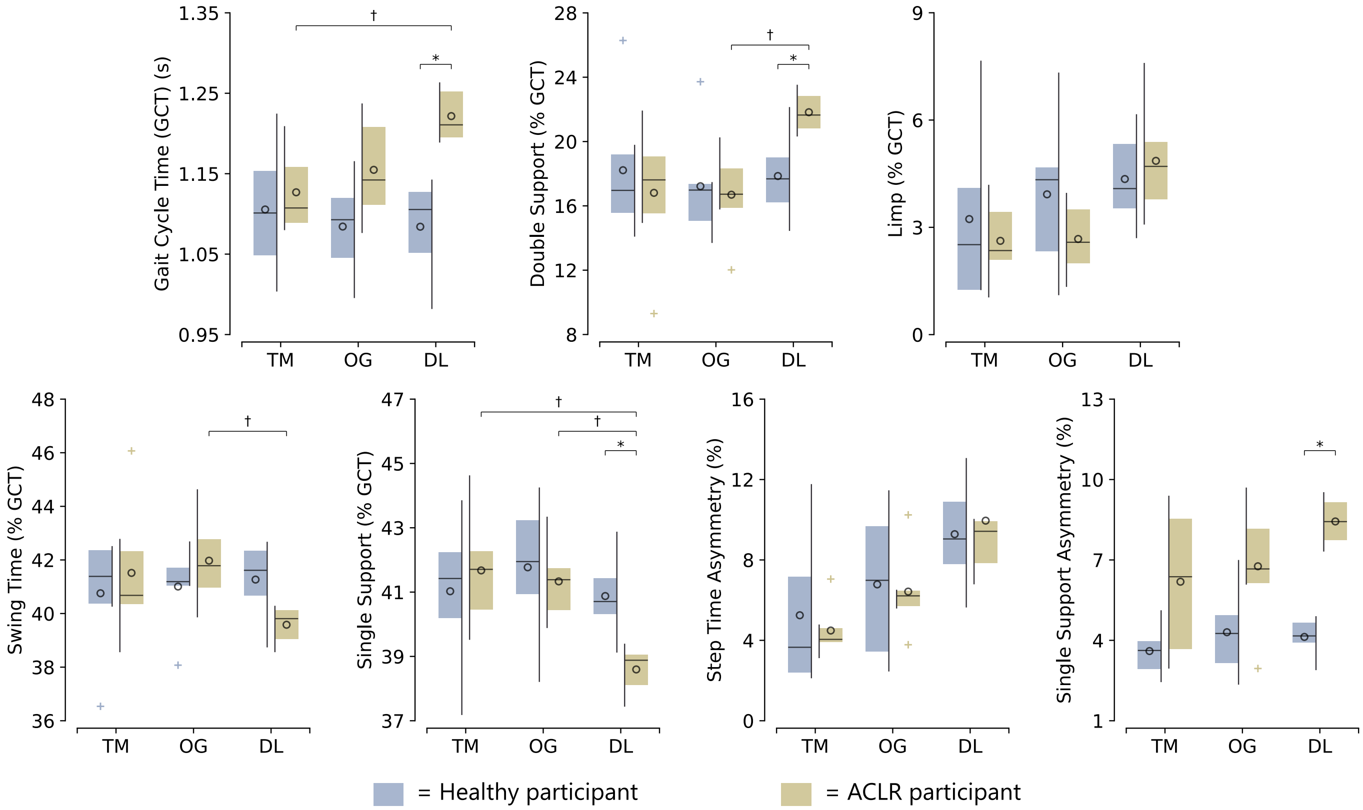

Fig. 2. Pairwise Post Hoc Comparisons. Patients post-ACLR walked with longer gait-cycle and double-support times and shorter single-support times in daily life compared to the laboratory. Patients walked similarly to healthy participants in laboratory conditions, but not during daily life. A dagger denotes statistical significance for post hoc pair wise comparisons related to hypothesis 1 (i.e., differences across environments for ACLR patients only). An asterisk denotes statistical significance for post hoc pair wise comparisons related to hypothesis 2 (i.e., differences between groups, across environments). All symbols to denote significance indicate $p < 0.017$. Plus sign markers denote outliers. The large bars represent the min-max range of the box plot (Q1 - 1.5* Interquartile range (IQR) to Q3 + 1.5*IQR). Circles within the box plots denote mean, while horizontal bars denote median. The top three plots focus on the three primary outcomes, while the bottom ones on related secondary outcomes.

10-20, 20-40, 40-80, 80-150, and 150+ gait cycles. For all three models, we used a Box-Cox correction to overcome any non-normality in all the models. Where differences were observed in our mixed model, we completed pairwise comparisons with Bonferroni correction for multiple comparisons. Last, we used a t-test to determine which group (ACLR, healthy) was more active, where activity was defined as the total cycles analyzed per day, and a two-way RM ANOVA to determine if walking-bout duration histograms for each group were significantly different, hypothesizing that ACLR patients are less active and walk in shorter bouts.

All the summary statistics in the tables of the Results section and the abstract are reported as means and standard deviations since the data were normally distributed (Shapiro-Wilk's $p > 0.99$).

III. Results

Patients post-ACLR walked with greater gait-cycle and double-support times in daily life compared to the laboratory ($p < 0.017$; Figure 2; Table III). Limp also trended toward being greater in daily life compared to the laboratory ($p < 0.05$), but after Bonferroni correction these differences were not significant ($p > 0.017$). Similarly, secondary outcomes trended toward differences across environments, except for single-support time, which is well correlated with double-support time, these differences were either not statistically or not practically significant. All the tests were sufficiently powered to detect differences if they existed, with post-hoc power analyses revealing that group sample sizes of 6 or fewer were sufficient (Table III). These statistically significant differences were greater than the error of IMU-reported outcomes compared to ground-truth optical motion capture (Table II).

Patients post-ACLR walked similarly to healthy participants in the laboratory, but not in daily life, where they walked with longer gait-cycle and double-support time and shorter single-support time (Figure 2). Single-support asymmetry was also higher in ACLR patients than in healthy participants during daily life. The statistically significant differences were larger than the error of IMU parameters compared to optical motion capture (Table II). No other significant differences were observed. While some of the tests were underpowered (Table IV), the observed effect sizes indicate that even if these tests were sufficiently powered, the differences would not be practically significant. For example, step-time asymmetry differences of 1% between the two populations are lower than the accuracy of the IMU-based estimation (Table II), and therefore powering the experiment with the required sample (Table IV) may not be sensible.

TABLE II
PRACTICAL SIGNIFICANCE OF DETECTED DIFFERENCES

Parameter	H1: Differences across Environment				H2: Differences between Groups in Daily Life		
	Treadmill MAE	Abs. Diff. DL/TM	Overground MAE	Abs. Diff. DL/OG	Treadmill MAE	Overground MAE	Grp. Diff. A/H
Gait Cycle Time (GCT) (s)	**0.02**	**0.09 (0.05)***	0.02	0.07 (0.04)	0.01	**0.02**	**0.14***
Double Support (% GCT)	5.13	5 (3.7)	**4.44**	**5.1 (2.5)***	3.6	**3.31**	**3.9***
Limp (% GCT)	2.28	2.2 (2.3)	2.04	2.2 (2)	2.56	2.55	0.5
Swing Time (% GCT)	3.12	2.3 (1.9)	2.77	2.4 (1.7)	2.18	2.23	1.7
Single Support (% GCT)	**2.7**	**3.1 (1.5)***	**2.39**	**2.7 (1)***	2.09	**1.97**	**2.3***
Step Time Asymmetry (%)	3.38	5.5 (4.2)	4.58	3.8 (4.7)	3.63	5.19	0.7
Single Support Asymmetry (%)	5.23	3 (2.1)	4.13	2.4 (1.3)	4.25	**4.01**	**4.3***

All the statistically significant pair-wise differences across environments (H1) or between groups (H2) were greater than the respective minimal detectable difference (MDC) for that respective IMU-based parameter. Here MDC is defined as the Mean Absolute Error (MAE) of the IMU-based parameters compared to ground-truth optical motion capture. Absolute Differences (Abs. Diff.) between Daily Life and Overground conditions (DL/OG), Daily Life and Treadmill conditions (DL/TM), and group differences (Grp. Diff.) between ACLR patients and Healthy participants (A/H) are bolded for the parameters that showed significant differences (*p < 0.017). All the summary statistics are reported as means and standard deviations since the data were normally distributed.

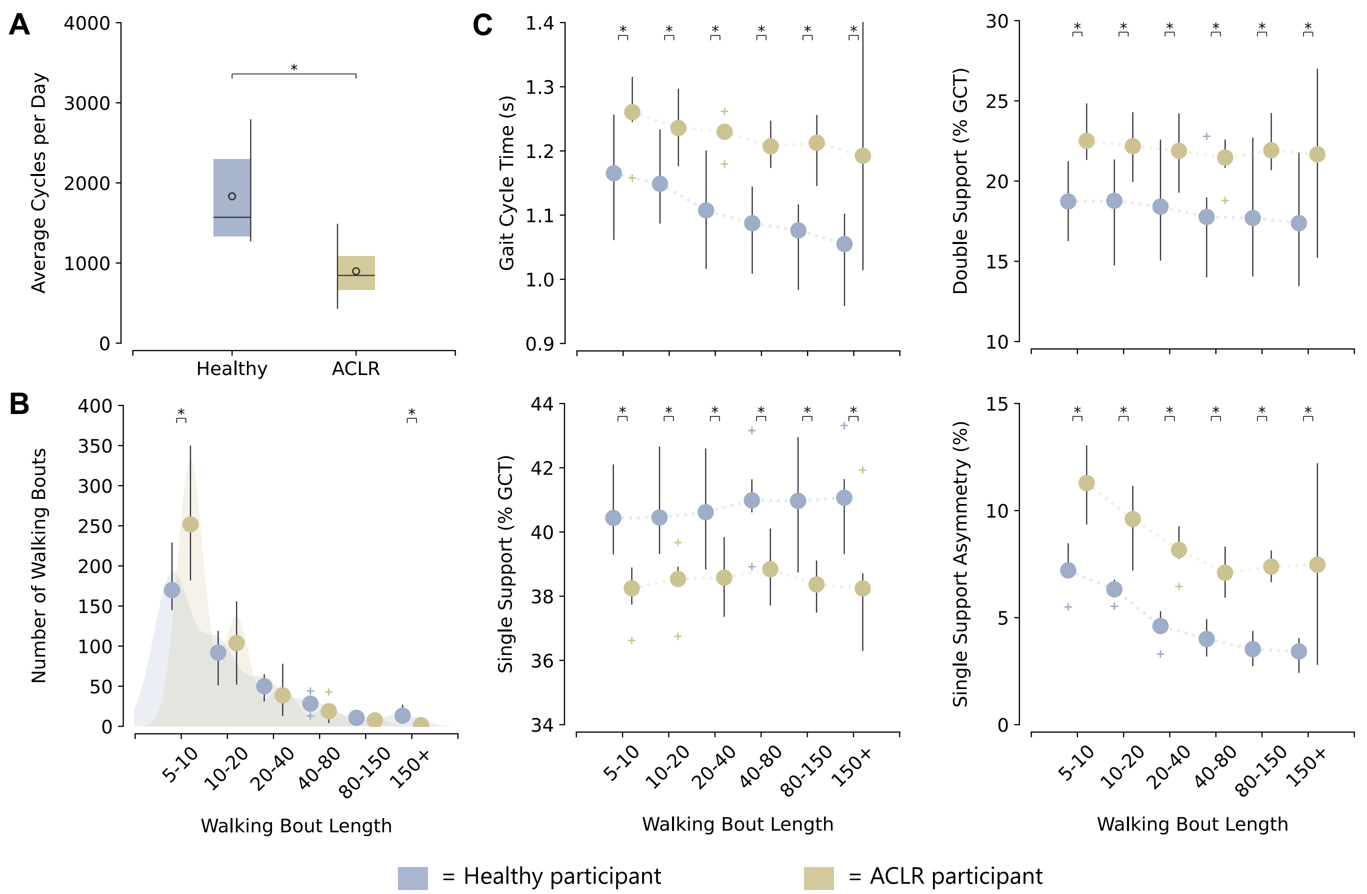

Fig. 3. Walking Bout Differences between Groups. A) Healthy participants completed more gait cycles per day than ACLR patients. The total gait cycle count only includes gait cycles belonging to walking bouts with 5 or more gait cycles. B) All participants walked predominantly with bouts of 5 – 10 gait cycles. Healthy participants, however, walked with longer bouts more often than ACLR patients. C) Double-support and single-support time (as % of gait-cycle time) remained unchanged across walking bout length. Gait-cycle time and single-support asymmetry decreased with walking bout length. Differences between healthy participants and ACLR patients persisted regardless of walking-bout length for all parameters. Single-support asymmetry is reported as a percentage representing the ratio of short single-support time over long single-support time per gait cycle. Plus sign markers denote outliers. The large bars represent the min-max range of the box plot (Q1 - 1.5* Interquartile range (IQR) to Q3 + 1.5* IQR). Circles within the box plots denote mean while horizontal bars denote median.

ACLR patients were less active and walked in shorter bouts compared to healthy participants during daily life. Healthy participants completed 935 more gait cycles per day than ACLR patients ($p = 0.016$)(Figure 3A). Both participant groups walked in predominantly short walking bouts (Cohen's $f = 3.716$, $p < 0.001$), but healthy participants had a higher portion of long bouts compared to ACLR participants (Cohen's $f = 0.670$, $p = 0.004$) (Figure 3B). Differences between ACLR and healthy participants persisted across walking-bout lengths for the following parameters: gait-cycle time (Cohen's

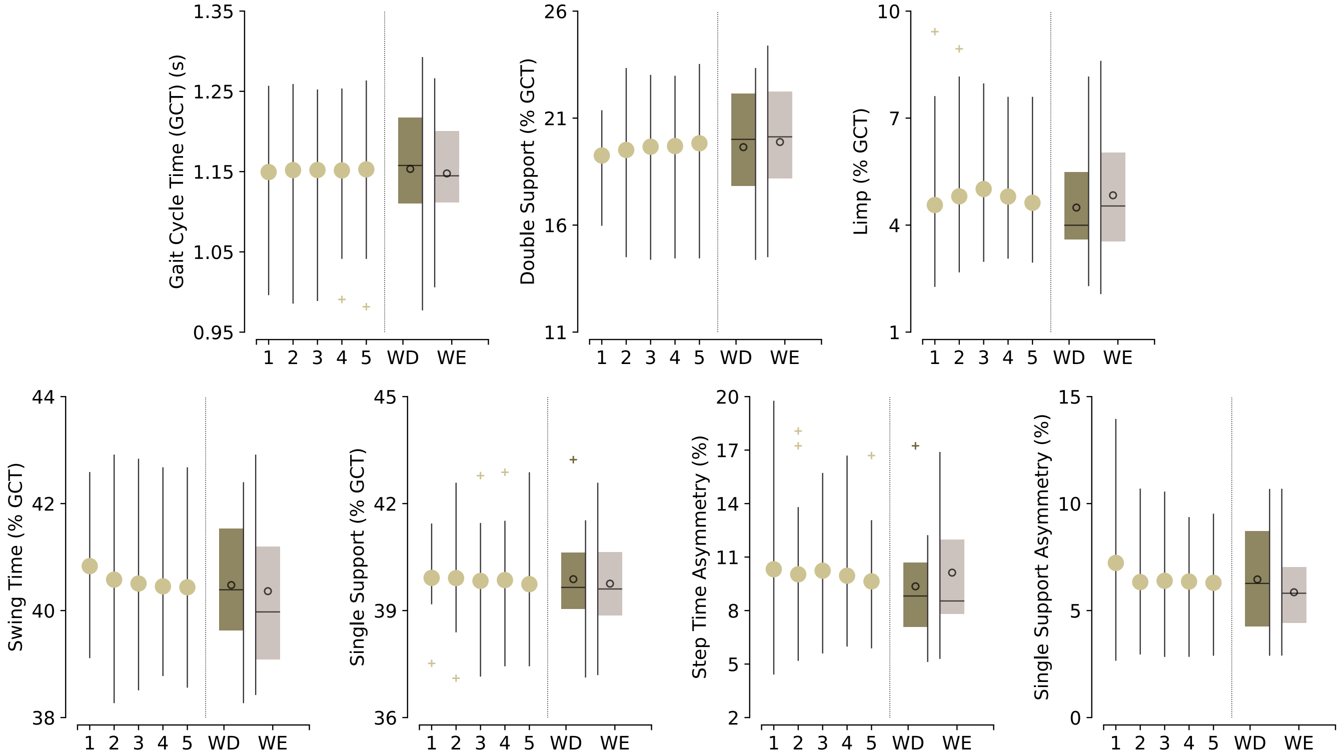

Fig. 4. Temporal Parameter Sensitivity to Day Type and Number. No significant differences were observed in any outcome when varying the number of days included in our analysis or selecting for weekday (WD) or weekend (WE) data. Asymmetry parameters are reported as a percentage representing the ratio of short parameter time to long parameter time per gait cycle. Plus sign markers denote outliers. The large bars represent the min-max range of the box plot (Q1 - 1.5* Interquartile range (IQR) to Q3 + 1.5* IQR). Circles within the box plots denote mean while horizontal bars denote median.

f = 1.07, $p = 0.003$), double-support time (Cohen's f = 0.87, $p = 0.004$), single-support time (Cohen's f = 1.07, $p = 0.003$), and single-support asymmetry (Cohen's f = 2.217, $p < 0.001$) (Figure 3C). Gait-cycle time (Cohen's f = 0.707, $p < 0.001$) and single-support asymmetry (Cohen's f = 2.23, $p < 0.001$) decreased with increasing walking bout length. Pairwise comparisons revealed that gait-cycle time only decreases with increasing walking bout length in healthy participants (Figure 3). The type of day (weekday or weekend) or number of days included in the analysis did not influence the outcomes(Figure 4).

IV. Discussion

This study suggests that a gait laboratory may not be the ideal environment to study post-ACLR gait and the impact it may have on long-term patient outcomes, especially PTOA. We found that patients following ACLR walk with longer gait-cycle times and rely more on double support in daily life compared to the laboratory. Interestingly, we also found that these patients walk similarly to healthy participants in the laboratory, but not in daily life, where they walk with longer gait-cycle time, longer double-support time, shorter single-support time, and greater single-support asymmetry. While we did not study the factors that make the gait laboratory not an ideal environment to capture free-living behavior, on their own, these findings provide initial evidence that the laboratory environment masks gait differences between healthy participants and ACLR patients and suggests that the move to natural-environment biomechanics holds new promise in expanding our understanding of post-surgical gait.

When interpreting these findings, it is important to consider the following characteristics of the study. First, healthy and ACLR groups were not age-matched. Gait matures by the age of seven [30], however, and prior work indicates that young adults and adolescents walk similarly in the context of temporal and asymmetry outcomes [31], [32]. Second, participants were tasked with applying the IMUs themselves each morning during the remote monitoring component of the study and no sensor-to-body calibration was implemented before extracting temporal parameters from the sagittal plane angular velocity data. While participants were provided visual aids through an app to optimize sensor alignment, human error is always a possibility. While validation of IMU-based daily-life outcomes is difficult to achieve, we report lab-based mean-absolute errors. Further, an independent analysis by our research group revealed that calibration of IMUs had no significant impact on temporal gait outcomes extracted with this approach when participants placed sensors by themselves. The effect of minor sensor misalignment seems to be significant on range of motion outcomes, but not on the temporal parameters reported here. A third limitation is the variable terrain in daily life compared to the laboratory. While we cannot conclude

TABLE III
HYPOTHESIS 1 TESTING AND POST-HOC POWER ANALYSES

Parameter	TM	OG	DL	p	SS	Power
Gait Cycle Time (GCT) (s)	1.13 (0.05)	1.15 (0.06)	1.22 (0.03)	0.0016*	3	0.88
Double Support (% GCT)	16.8 (4.3)	16.7 (2.8)	21.8 (1.3)	0.0086*	3	0.86
Limp (% GCT)	2.6 (1.2)	2.7 (1)	4.9 (1.6)	0.047	4	0.82
Swing Time (% GCT)	40.6 (2.6)	41 (1.7)	39.6 (0.7)	0.038	4	0.82
Single-Support (% GCT)	41.7 (1.8)	41.3 (1.3)	38.6 (0.8)	0.0015*	3	0.99
Step Time Asymmetry (%)	4.5 (1.4)	6.4 (2.1)	10 (3.5)	0.051	4	0.86
Single Support Asymmetry (%)	6.2 (2.8)	6.8 (2.3)	8.4 (0.9)	0.132	6	0.84

DL=Daily Life; OG=Overground; TM=Treadmill

Patients post- ACLR walked with higher gait cycle time and reliance on double support (primary) in daily life. Limp (primary) and secondary outcomes trended toward significant differences across environments. All outcomes were calculated from IMU data. Repeated Measures (RM) ANOVA post-hoc power analyses confirmed that this was not due to lack of power, with required sample sizes (SS) being 6 or less (*$p < 0.017$). All the summary statistics reported as means and standard deviations since the data were normally distributed.

with certainty the reason behind the differences between the laboratory and daily life outcomes, it suffices to know that the laboratory is not a realistic representation of natural environment gait. Although we could not control terrain, differences between patients and healthy participants were more pronounced in daily life than in the laboratory. We currently have no reason to believe that ACLR patients are moving in more challenging terrains than healthy participants. Fourth, a post-hoc power analysis for the second hypothesis revealed that limp and step time asymmetry required a larger sample size to help in rejecting the null. Yet, the observed differences between these parameters are lower than the accuracy with which these parameters can be estimated using IMUs (i.e., lower than the minimal detectable difference). It may therefore not be sensible to power a study for such small effect sizes that are competing with the accuracy of the measurement technology, at least until algorithmic advancements in IMU-based motion tracking match or surpass marker-based motion tracking. Last, previous work has demonstrated that recovery characteristics, such as quadriceps strength, may influence gait following ACLR. For example, it has been reported that patients with good quadriceps strength walk similarly to healthy adults, while those with weak quadriceps do not [33]. Clustering all ACLR recovery phenotypes into a single population is therefore not ideal, and modern data science tools such as cluster analysis and larger datasets will play a critical role in refining future analyses and insights.

Although previously unreported, the finding that ACLR patients walk differently in daily life than in the laboratory builds on the general consensus that the gait laboratory is not the ideal environment to study gait. Longitudinal gait analysis studies have reported that abnormal gait following surgery resolves over time [19], [34]. Laboratory studies report that temporal asymmetries resolve as early as three months following ACLR and that temporal parameters of gait do not differ between healthy and ACLR participants [19], [35], [36]. Our research corroborates these laboratory findings, observing no difference in temporal parameters of gait between ACLR and healthy participants in the laboratory (Figure 2). The differences we observed in temporal parameters between groups in daily life, however, suggest that what is observed in the laboratory may not be representative of movement patterns in daily life. This effect is mirrored in previous work in older adults [8], [10], [11], patients with Parkinson's disease [9] and multiple sclerosis [37]. Gait lab studies assess what is clinically known as a patient's capacity, which improves over time, but likely not daily life performance in their usual environment [38]. Functionally, a patient may have recovered enough to be able to walk symmetrically when asked to do so in a laboratory study, but they may still be adopting pain-avoidance strategies in daily life. Our results, together with the growing body of evidence in other clinical populations, therefore, call into question our existing understanding of gait recovery timelines, which have largely been informed by laboratory-based assessment and highlight the need for remote assessments to address the gait lab effect.

Additionally, the distraction-free laboratory environment may prompt patients to focus more on their perceived gait performance than in daily life. Typical laboratory data collection focuses on a single task, and the lab is devoid of any distractions, allowing participants to focus on walking alone. Gait in daily life, however, is rarely carried out in distraction-free environments. Previous work has demonstrated that patients following ACLR walk differently during dual-task compared to single-task walking [39]. Further, dual-task gait has been shown to be more correlated with daily life gait than single-task gait in older adults [6]. Gait features such as walking speed, knee joint underloading, and asymmetry have been shown to be indicative of long-term health outcomes and associated with poorer joint health following surgery [20], [40], [41], [42]. Distraction-free laboratory gait may inadvertently cause a participant to mask compensatory gait strategies in the laboratory. Therefore, remote monitoring could be the key to understanding the relationship between gait adaptation following ACLR surgery and long-term joint health.

As we move the study of gait to daily life, work that establishes direct correspondence between natural-environment and laboratory outcomes will be necessary to help place future findings in context with an extensive and valuable body of prior work. Here, we not only established how patients following ACLR move across different environments but examined these differences in relation to the precision that characterizes them (i.e., minimal detectable difference)

TABLE IV
HYPOTHESIS 2 RM ANOVA RESULTS AND POST-HOC POWER ANALYSIS

Parameter	Treadmill		Overground		Daily Life		Group X Environment		
	Healthy	ACLR	Healthy	ACLR	Healthy	ACLR	P value	SS	PW
Gait Cycle Time (GCT) (s)	1.11 (0.1)	1.13 (0.05)	1.08 (0.06)	1.15 (0.06)	1.08 (0.06)	1.22 (0.03)	0.0167*	8	0.92
Double Support (% GCT)	18.2 (4.4)	16.8 (4.3)	17.2 (3.5)	16.7 (2.8)	17.9 (2.7)	21.8 (1.3)	0.005*	6	0.90
Limp (% GCT)	3.2 (2.3)	2.6 (1.2)	3.9 (2.3)	2.7 (1)	4.4 (4.4)	4.9 (1.6)	0.3955	28	0.83
Swing Time (% GCT)	40.8 (2.3)	40.6 (2.6)	41 (1.6)	41 (1.7)	41.3 (1.5)	39.6 (0.7)	0.0146*	8	0.95
Single Support (% GCT)	41 (2.3)	41.7 (1.8)	41.8 (2.2)	41.3 (1.3)	40.9 (1.3)	38.6 (0.8)	0.004*	6	0.92
Step Time Asymmetry (%)	5.2 (3.9)	4.5 (1.4)	6.8 (3.8)	6.4 (2.1)	9.3 (2.7)	10 (3.5)	0.7802	112	0.81
Single Support Asymmetry (%)	3.6 (1)	6.2 (2.8)	4.3 (1.7)	6.8 (2.3)	4.1 (0.7)	8.4 (0.9)	0.2047	16	0.87

Gait differences between patients and healthy participants were dependent on the environment in which they were studied, with these differences being more pronounced in daily life than in the laboratory. As group differences were not observed in the laboratory environment, this table reports how group differences changed with environment. Of the primary outcome variables, only one (limp) did not contribute toward rejecting the null. SS denotes total required sample size. PW denotes achieved power. Data is reported as mean (standard deviation).

(Table II). While a few prior studies focusing on older adults [8], [10], [11] and patients with Parkinson's disease [9] and multiple sclerosis [37] have started to report differences between laboratory and daily-life gait, these deviations are different across populations and, most importantly, the accuracy with which a gait parameter can be captured with IMUs may also vary across populations.

The variability of bout length during daily-life gait may provide important information about post-ACLR gait that cannot be identified through laboratory gait assessments. Patients following ACLR were more sedentary and completed fewer long walking bouts per day than healthy participants, aligning with previous findings in ACLR patients (Figure 3) [43]. All participants in our study spent the majority of their day in shorter walking bouts, which is commonly seen in other populations [9], [37], [44], [45], [46]. Contradictory evidence has suggested that either long or short walking bouts are better suited for detecting differences between patients with Parkinson's disease and older adults [9], [46]. In our analysis, differences between groups persisted regardless of walking bout length (Figure 3). While gait-cycle time and single-support asymmetry decreased with increasing walking bout length, paired analyses revealed this gait-cycle time reduction was only observed in healthy participants. Population-specific gait changes with bout length have been identified in Parkinson's and OA gait, but detecting these compensatory adaptations requires datasets that capture the variability of bout length observed in daily life [9], [47]. While longer walking bouts may be more similar to typical laboratory gait trials, walking bouts of 30 seconds or less are reported to make up 60% of our daily activity [45]. These bouts bear clinical significance because they contribute to cumulative joint loading and excluding them would not be sensible in studies of orthopaedic populations.

Altogether, this work provides initial support for the use of multi-sensor protocols in remote patient monitoring following ACLR surgery. Ubiquitous monitoring after surgery could close the loop between research and clinical practice by granting researchers access to larger and more realistic datasets than they have been able to access to date in traditional laboratories and by allowing clinicians to easily incorporate the new insights drawn from these data into clinical practice. Even if traditional laboratory studies identified the gait patterns leading to poor recovery and eventual osteoarthritis, the gap between research and clinical practice would remain wide because traditional gait analysis is too expensive and time-consuming for implementation in clinics. Wearables are one of the few tools in the history of biomechanics that both researchers and clinicians are embracing with similar enthusiasm and hope for tangible impact.

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